

Emilio Hernandez-Martinez

Mechanical Engineering

PORTFOLIO

2024 – 2026

TABLE OF CONTENTS

<u>Profile</u>		1
<u>Vib-Iso Simulation Studies</u>		2
<u>Vib-Iso Engineering Drawings</u>		3
<u>Tempel End Effector</u>		4
<u>Vaisala Support Pivot</u>		5

PROFILE

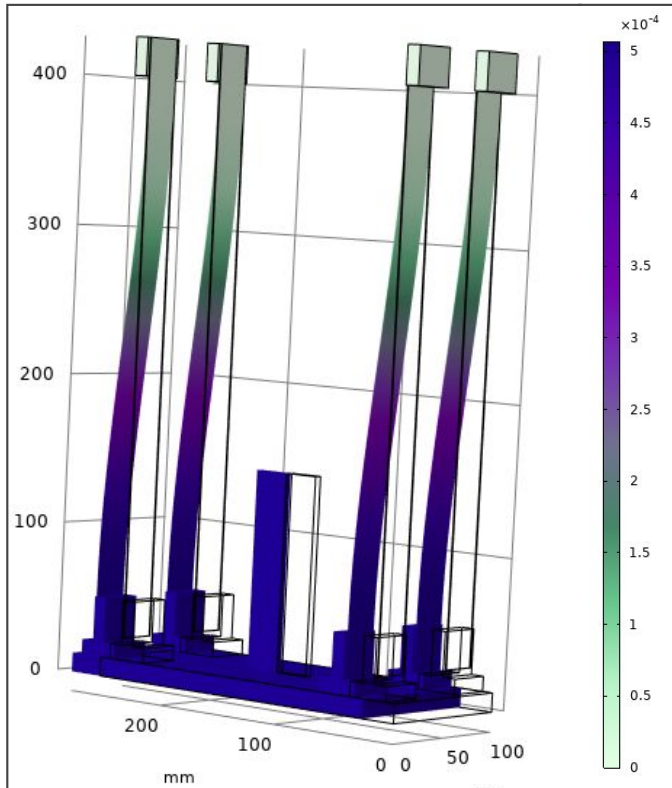


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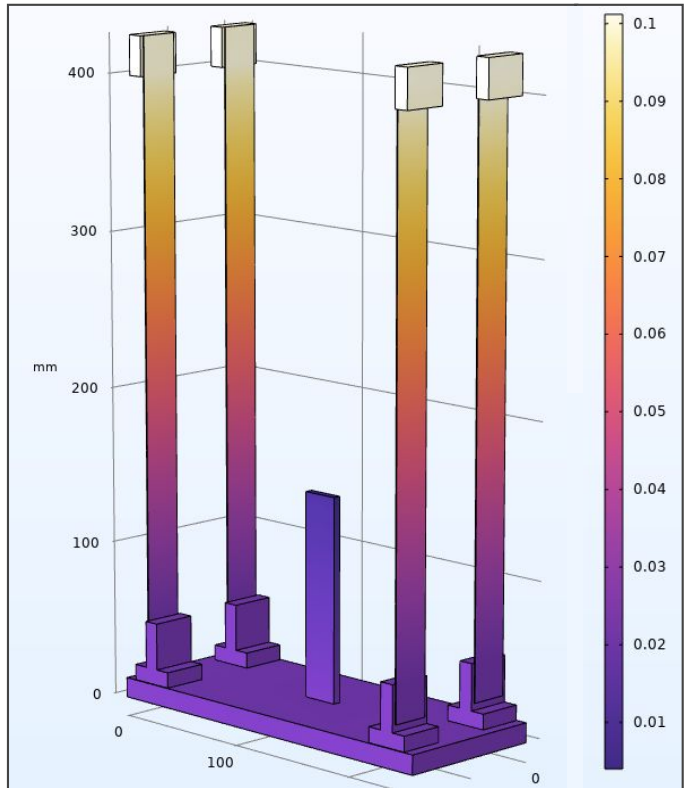
About

This portfolio highlights some of the engineering projects I've worked on through undergraduate research, mechanical design work, and competition robotics. As a mechanical engineering student, I value opportunities to apply technical concepts to engineering challenges involving simulation, analysis, design iteration, and real-world implementation. The projects included here span cryogenic vibration isolation research, COMSOL mechanical and thermal modeling, CAD development, and subsystem design, reflecting my continued growth in engineering problem solving, manufacturable design, and technical communication. Feel free to contact me with any questions about these projects through email.

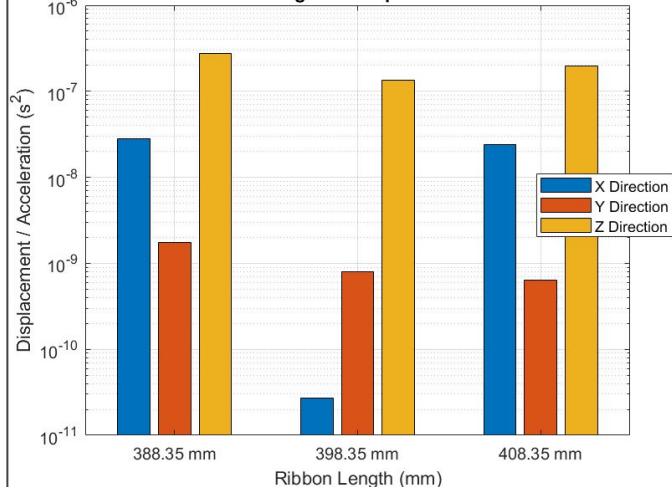
Cryogenic Vibration Isolation System (Vib-Iso) Simulation Studies

Frequency-Domain Displacement ($\text{mm} \times 10^{-4}$)

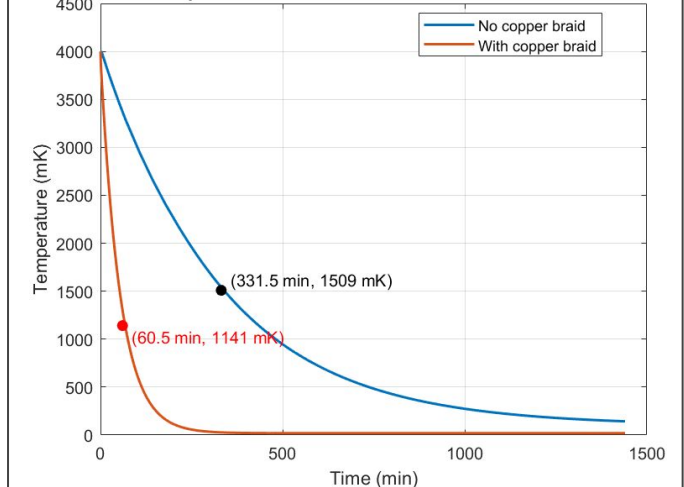
Volume Temperature Distribution (K)



Effect of Ribbon Length on Displacement / Acceleration



Temperature vs Time with Thermal Time Constants



Task: Designed a vibration isolation structure for a dilution refrigerator where both mechanical stability and thermal conduction are critical. Small changes significantly impact resonance behavior and cooling performance.

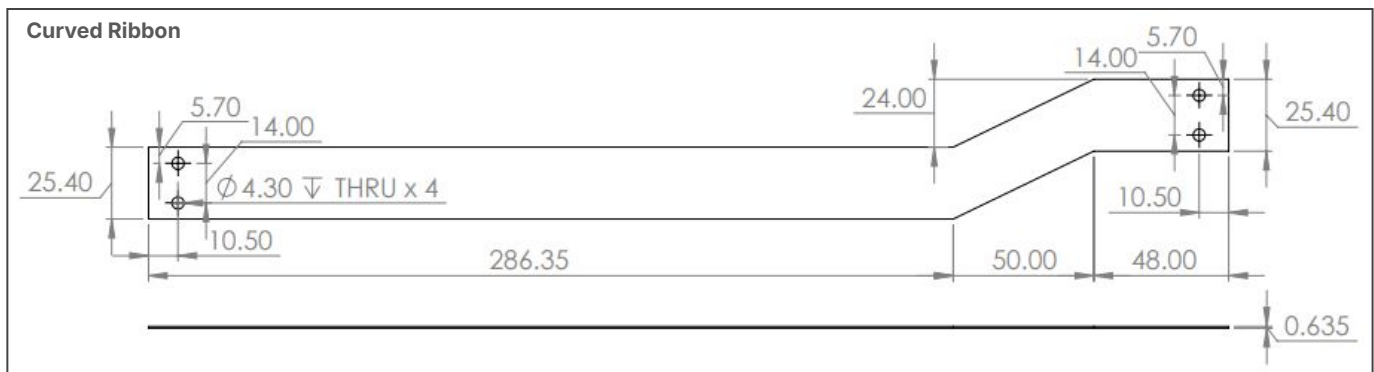
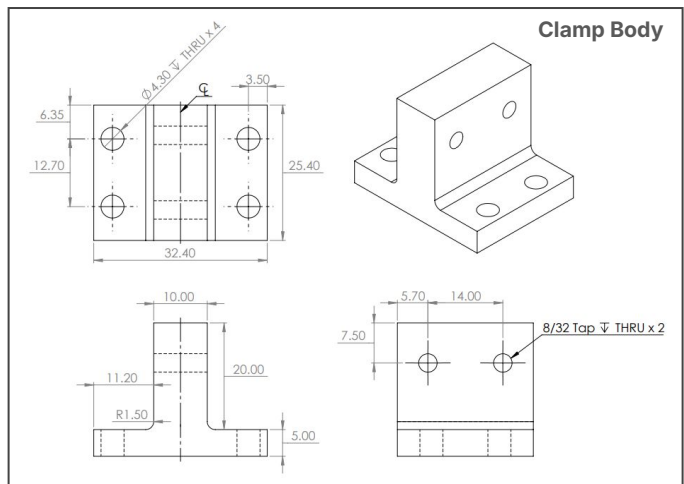
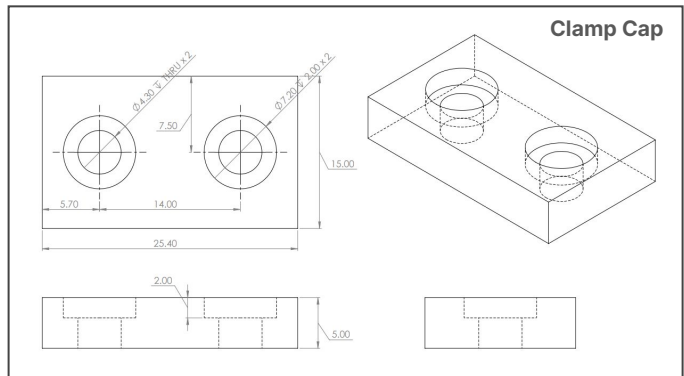
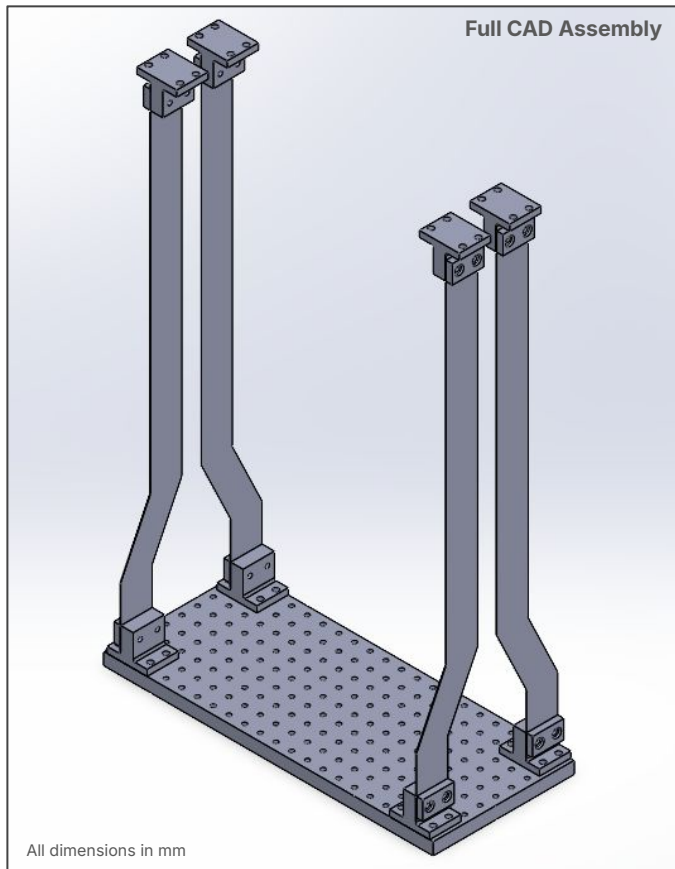
Approach & Key Insights

- Performed frequency-domain simulations under harmonic excitation
- Varied ribbon lengths (± 10 mm) to evaluate sensitivity
- ~398 mm ribbon length minimizes displacement

- Ran stationary and time-dependent thermal studies
- Compared designs with/without braid anchoring
- Copper braid significantly improves thermal conduction and cooldown

Impact: Enabled transition from simulation to fabrication-ready Vib-Iso system for cryogenic quantum experiments

Engineering Design & Drawings of Vibration Isolation System



Design Process:

- Converted simplified simulation model into detailed CAD
- Selected OFHC copper for high thermal conductivity
- New functional features: mounting holes, clamp geometry, etc.
- Designed curved ribbon to preserve optical line-of-sight

Manufacturability:

- Produced engineering drawings with tolerances and specs
- Incorporated threaded features (8-32, M3) for assembly
- Used section views and hidden lines to define internal geometry
- Optimized layouts to reduce ambiguity

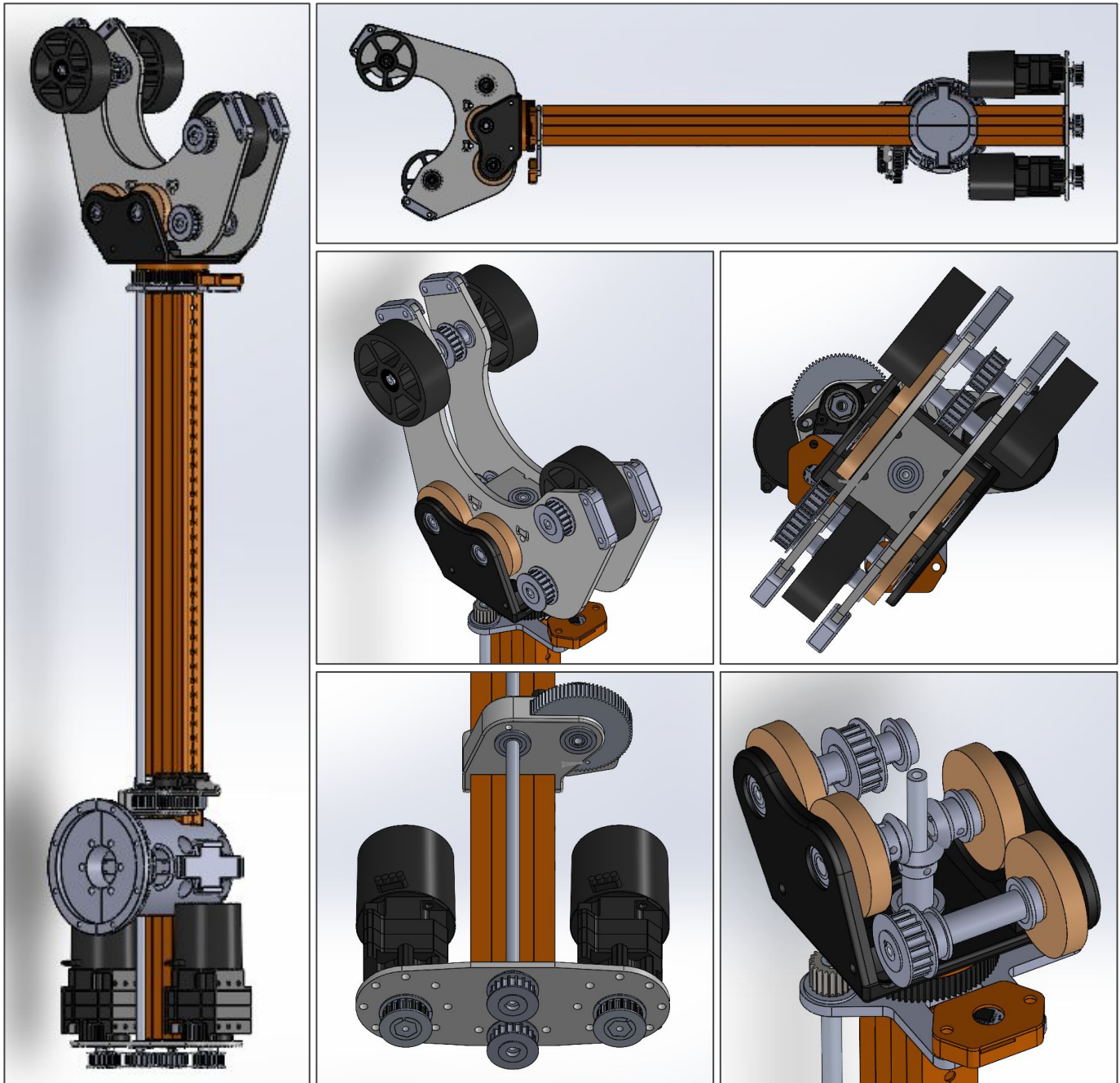
Challenges & Solutions:

- First drawing set: clear dimensioning and layout practices
- Resolved ambiguity in internal features using section views
- Reduced drawing clutter through iterative refinement
- Balanced design intent with fabrication constraints

Outcome: Completed engineering drawing set for all components; parts successfully manufactured and delivered

COMETS 2025 "Tempel"

Differential End Effector Subsystem



Objective: Design an end effector capable of complex game piece manipulation through independent wrist rotation and flywheel intake for FIRST Robotics competitions.

Key Components:

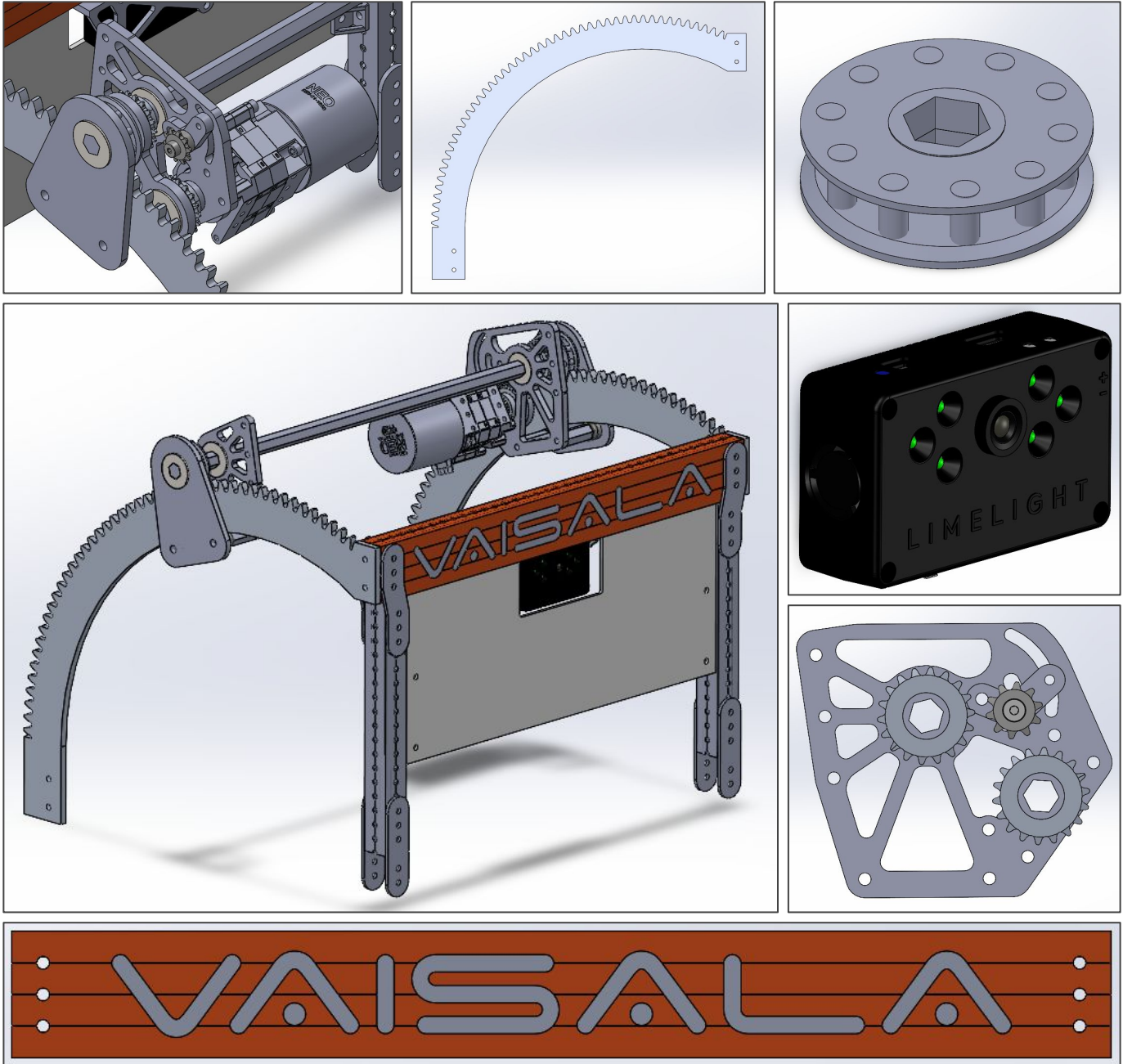
- Differential bevel gearbox, dual shafts for independent wrist and intake control
- Pulley-driven power transmission
- Modular assemblies for quick repair and integration
- Neo motors positioned opposite the wrist assembly for improved weight distribution

Challenges & Iteration:

- Intake geometry underwent multiple redesigns to optimize contact with game pieces
- Balanced rigidity, packaging, and rotational freedom
- Improved grip stability by transitioning from using four intake wheels to three
- Packaging required careful pulley and shaft routing

COMETS 2024 "Vaisala"

Dynamic Support Pivot Subsystem



Objective: Design a support pivot for adjusting the launch angle of the robot's shooting subsystem while maintaining drivetrain reliability, stiffness, and impact resistance during competition operation.

Key Components:

- Adjustable tensioning sprocket to maintain consistent tension across chain drivetrain
- Cross-shaft connected to dual pinion gears
- Large 0.25" polycarbonate gear arcs – high durability through repeated impact loads
- Limelight vision camera for automated targeting and robot odometry

Challenges & Iteration:

- Coordinating with other robot subsystems introduced packaging and alignment constraints
- Reduced mass of structural aluminum plates through lightweighting patterns
- Manufacturing large gear arcs required careful CNC workholding and multi-pass machining
- Balanced structural rigidity with manufacturability